ASSIGNMENT-4

Task – 1: Introduction to nano Editor

SOLUTION-

To learn the basic operations of the Nano text editor, including creating, saving, editing, searching, cutting, and pasting text.

Step 1: Create a File Using Nano

Command

nano nano_practice.txt

```
praveen@praveenasus:~$ nano nano_practice.txt
praveen@praveenasus:~$
```

Content Written in File

```
 GNU nano 8.7.1                    nano_practice.txt
Linux is an open-source operating system.
Linux was created by Linus Torvalds.
Linux supports multiple users.
Linux is known for its security.
Linux is widely used on servers.
Ubuntu is a popular Linux distribution.
Linux provides a powerful command line.
The Linux kernel manages hardware resources.
Linux is used in cloud computing.
Linux is stable and reliable.
```

Step 2: Save the File

Shortcut Used

Ctrl + O

Step 3: Exit Nano

Shortcut Used

Ctrl + X

Step 4: Search for a Word

Shortcut Used

Ctrl + W

Step 5: Cut a Line

Shortcut Used

Ctrl + K

Step 6: Paste a Line

Shortcut Used

Ctrl + U

Task – 2: vi / vim Modes Practice

SOLUTION-

To learn the basic modes and editing operations of the Vim editor.

➤Create a file using

vim vim_practice.txt

```
praveen@praveenasus:~$ vim vim_practice.txt
praveen@praveenasus:~$
```

Content written in file.

```
praveen@praveenasus: ~
Hlo
we are study about linux command vim and vi
~
~
~
```

Key Sequence-

- i = Enter Insert Mode
- Esc = Return to Normal Mode
- h j k l = Navigate cursor (move left , down , up , right)
- dd = Delete current line
- u = Undo last action
- Ctrl+R = Redo last action
- /Linux = Search for the word Linux
- :wq = Save file and quit Vim

Task – 3: vim vs nano Comparison

SOLUTION-

Vim

- Vim is a powerful text editor used in Linux.
- It works in different modes such as Normal Mode, Insert Mode, and Command Mode.
- It has a steep learning curve and requires practice.
- Vim uses the configuration file ~/.vimrc.
- It supports advanced search, replace, scripting, and syntax highlighting.
- Vim is best suited for advanced Linux users and system administrators.
- Vim: Difficult for beginners.

Nano

- Nano is a simple and user-friendly text editor.
- It works in a single editing mode.

- It is easy to learn and use.
- Nano uses the configuration file ~/.nanorc.
- It provides basic search and syntax highlighting features.
- Nano is best suited for beginners and simple text editing tasks.
- Nano: Easy to learn.

Task – 4: vim Configuration

SOLUTION-

To configure Vim using the .vimrc file and enable useful editing features.

Step 1: Open the .vimrc File

Command

vim ~/.vimrc

```
praveen@praveenasus:~$ vim ~/.vimrc
praveen@praveenasus:~$
```

Step 2: Add the Following Configuration

~/.vimrc Content

set number

syntax on

set tabstop=4

set shiftwidth=4

set autoindent

colorscheme desert

Explanation of Settings

set number

- Enables line numbers.
- Displays line numbers on the left side of the editor.

syntax on

- Enables syntax highlighting.
- Makes code and text easier to read.

set tabstop=4

- Sets the tab width to 4 spaces.

set shiftwidth=4

- Sets indentation width to 4 spaces.

set autoindent

- Automatically indents new lines.
- Maintains proper code formatting.

colorscheme desert

- Applies the "desert" color scheme.
- Improves editor appearance and readability.

Step 3: Save and Exit

Key Sequence

Esc

:wq

Enter

Step 4: Verify Configuration

Command

vim test.txt

Output -

1 Linux is an operating system.

2 Linux is open source.

3 Vim is a powerful editor.

Task – 5: Editing a System Configuration File

SOLUTION-

To edit a Linux system configuration file using Vim and verify the changes.

Step 1: View Current Content of /etc/hosts

Command

cat /etc/hosts

```
praveen@praveenasus:~$ cat /etc/hosts
# This file was automatically generated by WSL. To stop automatic generation o
f this file, add the following entry to /etc/wsl.conf:
# [network]
# generateHosts = false
127.0.0.1       localhost
127.0.1.1       praveenasus.localdomain praveenasus

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

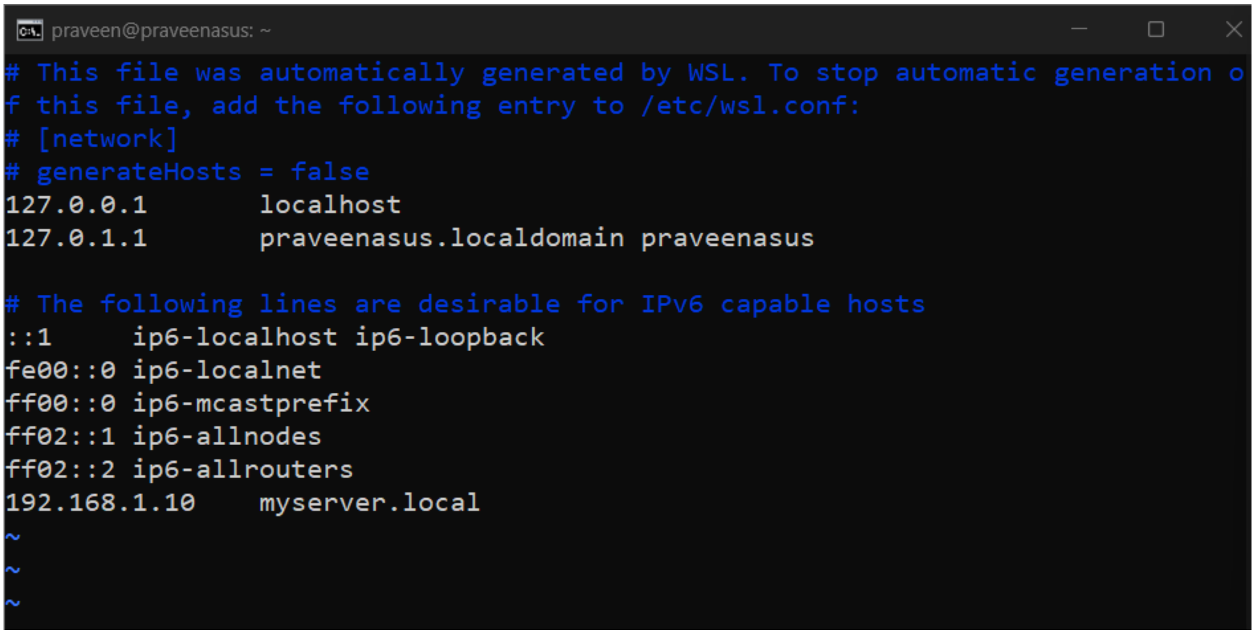
Step 2: Open the File Using Vim

Command

sudo vim /etc/hosts

```
praveen@praveenasus:~$ sudo vim /etc/hosts
[sudo: authenticate] Password:
```

Step 3: Add a Custom Hostname Entry



Add the following line:

192.168.1.10   myserver.local

Step 4: Verify the Change

Command

cat /etc/hosts